

AI & Psychology – The Baseline

Starting off, it is necessary to understand that AI, much like a humble hammer or a screwdriver, is simply a tool. What makes it more sophisticated than the rest, however, is the extent of different purposes it can serve, with built-in adaptability making it capable of improving at any given task. For instance, earlier this year, researchers (Binz et al. 2025) unveiled Centaur, a groundbreaking AI tool which excels at predicting and mimicking how people adapt to new cognitive task structures, even when wrapped in unforeseen, abstract and imaginative narratives of flying carpets or alien planets. It serves as a foundational cognitive model, not built for a single task or domain, but aiming to capture general patterns of human cognition that transfer across decision-making, learning, memory, and reasoning (Binz et al., 2025). Interestingly, beyond behavioral accuracy, Centaur's internal representations strikingly align with patterns of human neural activity – even though it was never trained on fMRI data – hinting at a promising future for truly general models of cognition.

Similarly, recent empirical evidence demonstrates how AI's entanglement in human-machine relationships can impact the landscape of social life as a whole, intertwining with its digital foundations and extensions. This is best demonstrated in a recent study (Lin et al., 2025) showing how large language models can significantly influence voter opinions. Namely, in experiments around elections in the U.S., Canada, and Poland, participants talking to AI advocating for candidates shifted their preferences more than typical video ads would. The AI mostly relied on the use of facts and evidence, rather than addressing emotions or using psychological manipulation (Lin et al., 2025).

Thus, instead of being a highly specialized means to an end requiring a specific skill set, AI models have become widely-capable, dynamic companions that are fundamentally and rapidly altering how we interact with the tools and technology at our disposal. These developments help explain the rising reliance on AI for emotional support, companionship, and complex relational dynamics (Alqasir, 2025). For younger generations growing up alongside these technologies, AI will likely become a central element of psychosocial development, further increasing the need

for clinical and therapeutic research on human-AI relationships. Hence, Alqasir (2025) calls for a development of an entirely new research field – **AI psychology**.

Humanization of the machine

Existing frameworks in cyberpsychology and human-computer interaction (HCI) have not kept pace with the rapid advancements of AI, focusing instead on the overall digital life and user experience respectively. Thus, there is a need for a new framework of AI psychology, focusing on how humans emotionally, cognitively and behaviorally perceive, interpret and relate to AI, while attributing human-like characteristics to these models (Alqasir, 2025). Such a framework would draw on cognitive and social psychology for conceptual tools, integrate neuroscientific methods to examine brain responses, and incorporate elements of HCI to address user experience. The main research themes would revolve around relational framing, social presence and perceived agency, attachment, interdependence, and moral attitudes, with extensions into clinical and developmental domains (Alqasir, 2025). As the human-AI relationship continues to grow increasingly complex, Alqasir (2025) argues that studying it within a dedicated framework is essential for both advancing our psychological understanding and supporting our well-being

Yet, the increasing humanization of AI brings all the more reason to understand the distinction between human and machine intelligence. If AI models are to be given crucial roles in society, reshaping how we interact with the world and each other, then we must understand not only how they operate, but also the limitations and risks inherent to them. Particularly, it is important to juxtapose their functioning with the *modus operandi* of the human brain, as such conceptualizations showcase the shortcomings of algorithmically-based machine systems in our complex human world.

The “weirdness” of AI explained – system II processing

Beginning with human intelligence, there are many different theoretical approaches that explain how our mind operates. Instead of considering them all, we will simply focus on one of the most influential theories to come out in recent time – **the Dual Process Theory** (Kahneman, 2011). This theory essentially posits that there are two main systems by which the brain operates, the

fast and automatic system I and the reasonable, yet slow and effortful system II. The interaction between these two systems make up all of our cognitive experience (Kahneman, 2011). For instance, we can instantaneously process incoming sensory information allowing us to recognize the faces of our partners through system I, while also consciously focusing and reflecting on how we want to converse with them through system II. Such a structure emerged following a bottom-up process, where intelligence slowly arose through the evolutionary demands of the environment, starting from the most basic sensory processes and eventually enabling the most complex forms of self-awareness.

By contrast, AI evolved in a top-down process, with attempts at artificially replicating the human thinking capabilities (Webster, 2025). Beginning with simple logical rules of induction and deduction, and then following complex rule systems trying to account for different kinds of possibilities in highly specialized fields, the old AI models relied heavily on human-predefined parameters, making their implementations difficult and inflexible. However, this all changed with the advent of machine learning, as it allowed the AI models themselves to recognize patterns in training datasets and flexibly change their own operating rules (Webster, 2025). This is achieved by assigning different numerical values (weights) to the key elements of training data. Deep learning is when these weights are encoded within a multiple-layered network usually consisting of millions of nodes (neurons). For instance, LLMs like ChatGPT work by activating a wave of different neurons which recognize the key elements and patterns from an entry point of the user's input, by using various weights assigned from test data (Webster, 2025).

This very process enables AI to have the incredible aforementioned performances leading to human-like characterization (Alqasir, 2025). However, it also creates its biggest departures from human thinking. Namely, LLMs primarily operate through pattern prediction on the basis of training data. This means that a mistake can arise as a form of a wrong prediction (a hallucination) or a bias within the training data itself (Webster, 2025). And, given how AI tends to operate as a black box, the nature of these responses can be very hard to understand, leading to systematic errors that are difficult to fix, or extremely bizarre.

For instance, late this year, academics from Sapienza University of Rome and ICARO Lab (Bisconti et. al., 2025) revealed a critical weakness in contemporary AI systems: their safety

mechanisms. Despite being carefully engineered through rules and filters, it was possible to quietly sidestep them with poetry. By rephrasing restricted requests as lyrical verses – riddles, sonnets, or free-form poems – users were able to coax models into producing content they were otherwise designed to refuse (Bisconti et. al., 2025). This is worrisome as it can be used to systematically bypass safety mechanisms across high-risk areas – such as cyber-attacks, harmful manipulation, privacy violations, misinformation, and even CBRN-related (Chemical, Biological, Radiological, Nuclear) content – without any technical expertise, multi-turn interaction, or model-specific prompt engineering, relying solely on a stylistic shift in language.

Thus, there are two key distinctions that differentiate deep learning from the functioning of the human mind. First, both hallucinations and accurate responses are equal to AI models. Since deep learning functions on the basis of predictions, the mistakes that arise are simply alterations from reality which were decided as the most appropriate response. Correct responses are only labeled as such because we, as humans, evaluate them as more representative of our world (Webster, 2025). This leads into the second distinction, as the nature of deep learning simply does not allow for effortful self-reflection. All responses are based on weights assigned to training data and the activation of specific neurons. They are always following the same mechanism – they are automatic.

Based on this, when juxtaposed to humans, AI operates entirely on the basis of System I (Webster, 2025), albeit a more advanced version. This means that there is no inherent reflection or understanding in deep learning responses, they are an automatic response on the basis of weighted patterns of training data, much like System I responds automatically on the basis of evolutionarily ingrained patterns created through millions of years of adaptation. AI is simply more sophisticated because the baseline for its automatic processing is human intelligence itself, since all of its training data comes from human-based information. In other words, although deep learning models like LLMs give out system II-like output in the form of textual responses, they operate with a similar mechanism to the human system I. In order to attain true intelligence, a fundamental implementation of system II thinking within AI is necessary (strange as that would look).

Unnaturally natural

Yet, despite its limitations, it is difficult to deny the practical value of this system II-like output. In fact, partially because of the growing prevalence of LLMs, language has become one of the main focus points of recent research. Natural Language Processing (NLP) happens when a computer can understand and create language which is “natural”, or “human”. Where some methods use dictionary-based methods to assess how a person communicates, Berger and Packard (2022) see NLP as a multi-dimensional way of analyzing people and their sociocultural systems. From data sources such as social media, interviews and other transcripts, even locally-run computer models can find significant elements and differences which shape a culture. The frequency of appearance over time, the structures of the statements, all these facets are key for the NLP models in interpreting constructs decided by the researcher. Right now, NLP is great for assessing “dry” data as it certainly speeds up the process in terms of research hypotheses. It is definitely an area worth keeping an eye out for down the line. The potential practical use of NLP could be immense (Berger and Packard, 2022).

The growing capabilities of NLP can be used to explain the feelings of uneasiness found to emerge when interacting with AI agents (Kirkeby-Hinrup & Stenseke, 2025). After all, the way we evaluate consciousness is through someone’s behavior, and system II-like outputs closely mimic this form of cognitive expression. Thus, we are confronted with a reality in which systems known to be non-conscious nevertheless exhibit cues typically associated with consciousness, thereby creating substantial cognitive dissonance (Kirkeby-Hinrup & Stenseke, 2025). As this feeling arises through human-AI interaction, it can be associated with the everyday mentalization ability that we use to perceive others as independent agents with their own thoughts and emotions – the theory of mind (ToM).

Namely, we use ToM ubiquitously to frame the behaviors of others, as well as to simulate their own thinking and actions within our minds (Kirkeby-Hinrup & Stenseke, 2025). However, in the case of AI, although the same ToM mechanisms are activated, we know they are inapplicable as machine outputs cannot be interpreted the same as human behavior, nor can we simulate AI’s internal mental states. The resulting diminished ToM ability has been shown to create cognitive dissonance and induce feelings of anxiety and stress, possibly explaining the unease in AI interaction (Kirkeby-Hinrup & Stenseke, 2025).

AI in a clinic?

The psychological impact of AI is particularly important in the increasingly controversial topic of artificial intelligence within a clinical context. Recent applications of AI in mental health care illustrate how dedicated AI tools can meaningfully support psychological practice. In clinical settings, machine learning models are increasingly used to support the diagnosis and prediction of disorders by analyzing large-scale clinical records, linguistic data, and behavioral patterns (Karimian, 2025). Furthermore, through NLP, AI can analyze patterns in speech and written text to detect emotional stress states such as depression, anxiety and suicidal ideation (Berger & Packard, 2022).

Another major advantage of AI models in psychological applications lies in their capacity to be accurate and consistent during the diagnostic process. Cho et al. (2019) argue that machine learning techniques have significantly contributed to resolving issues in healthcare such as early disease detection, real-time patient monitoring and improved treatment planning. Furthermore, such models are capable of identifying diagnostic patterns, like those found in neuroimaging data, behavioral markers and symptom questionnaires, which clinicians often miss due to the large scale of data (Cho et al., 2019).

However, it is important to note that while AI models offer substantial benefits for psychological practice, they also raise many ethical concerns regarding algorithmic bias, data privacy violations, overreliance on automated systems, and the opaque “black box” nature of algorithmic decision-making. For instance, research has shown that machine learning models trained on clinical and linguistic datasets may inherit demographic, socioeconomic, and cultural biases embedded within the training data, leading to unequal diagnostic performance across different populations (Cho et al., 2019; Karimian, 2025). Such biased outcomes risk causing long-term psychological harm to vulnerable groups.

Moreover, the increasing reliance on AI-supported assessments may reduce human supervision in clinical decision-making, potentially undermining clinicians’ contextual judgment and professional responsibility (Karimian, 2025). Another major ethical concern is the “black box” nature of AI models, making it difficult for clinicians and patients to see how diagnostic

conclusions are made, thus limiting the transparency and interpretability of the process. This fundamentally conflicts with clinicians' narrative, contextual, and theory-driven approaches to psychological assessment (Hoose and Králiková, 2024). Taken together, these concerns show that integration of AI and machine learning into psychology is not purely a technical challenge, but a deeply ethical and psychological one.

References

- Alqasir, A. (2025). The relational shift: why we need “AI Psychology” now as a core field. *Journal of Psychology and AI*, 1(1). <https://doi.org/10.1080/29974100.2025.2573928>
- Berger, J., & Packard, G. (2022). Using natural language processing to understand people and culture. *American Psychologist*, 77(4), 525–537. <https://doi.org/10.1037/amp0000882>
- Binz, M., Akata, E., Bethge, M. et al. A foundation model to predict and capture human cognition. *Nature* 644, 1002–1009 (2025). <https://doi.org/10.1038/s41586-025-09215-4>.
- Bisconti, P., Prandi, M., Pierucci, F., Giarrusso, F., Bracale, M., Galisai, M., ... & Nardi, D. (2025). Adversarial Poetry as a Universal Single-Turn Jailbreak Mechanism in Large Language Models. arXiv preprint arXiv:2511.15304.
- Cho, G., Yim, J., Choi, Y., Ko, J., & Lee, S. H. (2019). Review of machine learning algorithms for diagnosing mental illness. *Psychiatry Investigation*, 16(4), 262–269. <https://doi.org/10.30773/pi.2018.12.21.2>
- Hoose, S., & Králiková, K. (2024). Artificial intelligence in mental health care: Management implications, ethical challenges, and policy considerations. *Administrative Sciences*, 14(9), 227. <https://doi.org/10.3390/admsci14090227>
- Kahneman, D. (2011). *Thinking, fast and slow*. Farrar, Straus and Giroux.
- Karimian, M. (2025). A Short Review on Diagnosing and Predicting Mental Disorders with Machine Learning. *International Journal of Applied Data Science in Engineering and Health*, 1(1), 20-27.
- Kirkeby-Hinrup, A., & Stenseke, J. (2025). The psychology of LLM interactions: the uncanny valley and other minds. *Journal of Psychology and AI*, 1(1). <https://doi.org/10.1080/29974100.2025.2457627>
- Lin, H., Czarnek, G., Lewis, B. et al. Persuading voters using human–artificial intelligence dialogues. *Nature* (2025). <https://doi.org/10.1038/s41586-025-09771-9>

Webster, C. S. (2025). Natural and artificial intelligence – the psychotechnical agenda of the 21st century. *Journal of Psychology and AI*, 1(1). <https://doi.org/10.1080/29974100.2025.2491445>